This implies, $\Delta x = \tan(y + \Delta y) - \tan y$

$$\text{Therefore, } \frac{\Delta y}{\Delta x} = \frac{\Delta y}{\tan(y+\Delta y) - \tan y}$$

$$= \frac{1}{\dfrac{\tan(y+\Delta y) - \tan y}{\Delta y}}.$$

As $\Delta x \to 0$, $\Delta y \to 0$ also, so that

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{1}{\lim\limits_{\Delta y \to 0} \dfrac{\tan(y+\Delta y) - \tan y}{\Delta y}} = \frac{1}{\dfrac{d}{dy}(\tan y)}$$

$$= \frac{1}{\sec^2 y}$$

$$= \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$$

That is, $\boxed{\dfrac{d}{dx}(\tan^{-1} x) = \dfrac{1}{1+x^2}}$.

(iv) The derivative of arccot x or $\cot^{-1} x$

We know the identity

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}.$$

$$\text{This implies, } \frac{d}{dx}(\tan^{-1} x + \cot^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2}\right) = 0$$

$$\text{This implies, } \frac{d}{dx}(\tan^{-1} x) + \frac{d}{dx}(\cot^{-1} x) = 0$$

$$\text{That is, } \frac{d}{dx}(\cot^{-1} x) = -\frac{d}{dx}(\tan^{-1} x) = -\frac{1}{1+x^2}$$

That is, $\boxed{\dfrac{d}{dx}(\cot^{-1} x) = -\dfrac{1}{1+x^2}}$.

(v) The derivative of arcsec x or $\boxed{\dfrac{d}{dx}(\sec^{-1} x) = \dfrac{1}{x\sqrt{x^2-1}}}$.

(vi) The derivative of arccosec x or $\boxed{\dfrac{d}{dx}(\operatorname{cosec}^{-1} x) = \dfrac{-1}{x\sqrt{x^2-1}}}$.

The proofs of (v) and (vi) are left as exercises.

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